

Fengyan Zhang

Geospatial & Data



Contact

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Delft, The Netherlands

Education

Master's Degree

[TU Delft](#) | [Geomatics](#)

Thesis: [Snap rounding polygons with a triangulation](#)

Sep. 2021 – Jun. 2023

Bachelor's Degree

[Southeast University \(Project 985\)](#)

Geographic Information Science

Thesis: *Pedestrian travel trajectory generation based on Fréchet distance*

Sep. 2017 – Jun. 2021

Exchange Program

[University of Minnesota](#)

Geographic Information Science,

transport planning and Cartography

Jul. 2019 – Aug. 2019

Technical & Soft Skills

C++ / Python / JavaScript / SQL

[CGAL](#) (Computational Geometry),

[GDAL](#) (Geospatial Data), [LASTools](#)

(point cloud), [nlohmann-json](#) (JSON

for modern C++), CMake, Linux

(Ubuntu, WSL), LaTeX, QGIS, ArcGIS,

PostgreSQL/PostGIS,

English (working proficiency), Dutch

(beginner), German (beginner)

Passions

Snowboarding, piano, jogging
reading, writing

Summary

Former Research Assistant at [RWTH Aachen](#) and MSc graduate in Geomatics from [TU Delft](#). I work with software and data systems, with experience in Python and C++ for building data processing pipelines, geometric algorithms, and distributed sensor-based applications. My work has involved system integration, real-time data ingestion, and applied research in spatial and urban contexts. I enjoy working with a structured and pragmatic approach to problem-solving, focusing on reliability, clarity, and maintainability. I am interested in roles where engineering, data, and domain knowledge come together in real-world systems.

Professional Experience – Research Assistant

[Geoinformation and Geodetic Institute](#), RWTH Aachen | Nov. 2023 – Oct. 2025

- Developed and maintained interfaces for real-time data transmission and visualization via [OGC SensorThings API](#) and [MQTT](#) protocol; authored a conference paper: [Real-Time Sensor Data Integration for BIM-Based Hydraulic Structure](#).
- Integrated monitoring data into a Building Information Modeling (BIM) environment using linked data models.
- Collaborated with industry partners to refine workflows and optimize software extensions for hydraulic structure monitoring.
- Integrated [OGC API – Processes](#) into web portals.
- Supported teaching activities in Distributed & Web GIS and Geodatabase courses, prepared exam questions, exercises, and solutions.

Relevant Projects (Public Access)

- [snapoly](#): Algorithm implementation of my master thesis: [Snap rounding polygons with a triangulation](#).
- [CityJSON](#): Implementation of calculating building volumes, number of building floors, area, and orientation of roof surfaces.
- [geoCFD](#): Preprocess the geometry for CFD simulation - remove internal faces between adjacent buildings. This is developed as the [Nef Polyhedra](#) method for the MSc Geomatics Synthesis Project [facesBgone](#).
- [BIMConvertToGeo](#): Convert a building information modeling (BIM) model, specifically an IFC file, into a CityJSON file.
- [Reconstruct 3D Geometry](#): 3D geometry reconstruction based on the open-source project - [Easy3D](#).
- [LCP Runoff modeling](#): Implementation of the least cost path algorithm (LCP) for calculating flow direction and flow accumulation.
- [Spatial interpolation](#): Implementation of Nearest Neighbor (NN) / Inverse Distance Weighting (IDW) / Triangulated Irregular Network (TIN) and Laplace Interpolation.

Extracurricular

Operation Management Department Intern, Radiance Group. Jul. – Sep. 2020.